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VEHICLE FAULT PREDICTION SYSTEM v2.0
Two-Level Diagnostic Architecture
Based on SAE J1979 Standard

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[1/4] Generating training dataset from SAE J1979 thresholds...
Generated 4800 training samples (7 fault classes + normal)
[2/4] Training neural network (MLP 12->32->16->8)...
Training complete. Best epoch: 92
Test Accuracy: 82.29%

[3/4] System ready.
[4/4] Starting interactive interface...

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[ORANGE] Mass Air Flow Rate
Normal value : 8.00 g/s
New value : 39.91 g/s (+398.9%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Short Term Fuel Trim: 0.00 %
OK Fuel Tank Pressure: 0.00 kPa
OK Throttle Position: 20.00 %
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

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DIAGNOSIS REPORT – RANDOM SIMULATION

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Mass Air Flow Rate
Value: 39.91 g/s > ORANGE threshold (35.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------|-------|
| 1. Air Intake Fault | 49.8% |
| 2. Engine Overheating | 48.3% |
| 3. No Fault | 1.9% |

FINAL DIAGNOSIS

STATUS : ANOMALY DETECTED (fault type unclear)
ALERT : MEDIUM [ORANGE]
CONFIDENCE : 49.8% (below threshold for fault ID)
ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Mass Air Flow Rate: 39.91 g/s

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 3 parameter(s)...

CHANGES MADE:

[RED] Vehicle Speed

Normal value : 60.00 km/h
New value : 184.13 km/h (+206.9%)

[RED] Battery Voltage

Normal value : 14.00 V
New value : 10.43 V (-25.5%)

[ORANGE] Component Wear Index

Normal value : 50.00 min
New value : 224.71 min (+349.4%)

Parameters NOT changed (remain normal):

- OK Engine Coolant Temperature: 92.00 C
- OK Engine RPM: 1500.00 rpm
- OK Engine Load: 35.00 %
- OK Intake Air Temperature: 25.00 C
- OK Short Term Fuel Trim: 0.00 %
- OK Fuel Tank Pressure: 0.00 kPa
- OK Mass Air Flow Rate: 8.00 g/s
- OK Throttle Position: 20.00 %

OK 02 Sensor Voltage: 0.45 V

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DIAGNOSIS REPORT – RANDOM SIMULATION

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Battery Voltage

Value: 10.43 V << RED threshold (11.5)

[HIGH [RED]] Vehicle Speed

Value: 184.13 km/h >> RED threshold (180.0)

[MEDIUM [ORANGE]] Component Wear Index

Value: 224.71 min > ORANGE threshold (200.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|---------------------------------|-------|
| 1. Electrical / Charging Fault | 45.1% |
| 2. Transmission / Speed Anomaly | 40.8% |
| 3. Engine Mechanical Stress | 13.0% |

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FINAL DIAGNOSIS

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STATUS : ANOMALY DETECTED (fault type unclear)

ALERT : HIGH [RED]

CONFIDENCE : 45.1% (below threshold for fault ID)

ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Battery Voltage: 10.43 V
 - Vehicle Speed: 184.13 km/h
 - Component Wear Index: 224.71 min
- =====

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 22

Invalid option. Please enter 1, 2, or 3.

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 3 parameter(s)...

CHANGES MADE:

[RED] Engine Load

Normal value : 35.00 %

New value : 108.32 % (+209.5%)

[RED] Engine Coolant Temperature

Normal value : 92.00 C

New value : 135.64 C (+47.4%)

[YELLOW] O2 Sensor Voltage

Normal value : 0.45 V

New value : 0.03 V (-92.6%)

Parameters NOT changed (remain normal):

OK Engine RPM: 1500.00 rpm

OK Intake Air Temperature: 25.00 C

OK Short Term Fuel Trim: 0.00 %

OK Fuel Tank Pressure: 0.00 kPa

OK Mass Air Flow Rate: 8.00 g/s

OK Throttle Position: 20.00 %

OK Battery Voltage: 14.00 V

OK Vehicle Speed: 60.00 km/h

OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Engine Coolant Temperature

Value: 135.64 C >> RED threshold (120.0)

[HIGH [RED]] Engine Load

Value: 108.32 % >> RED threshold (95.0)

[LOW [YELLOW]] O2 Sensor Voltage

Value: 0.03 V < YELLOW threshold (0.1)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------------|-------|
| 1. Engine Overheating | 75.2% |
| 2. Engine Mechanical Stress | 12.2% |
| 3. No Fault | 12.2% |

FINAL DIAGNOSIS

STATUS : FAULT DETECTED

FAULT : Engine Overheating

CONFIDENCE : 75.2%
ALERT : HIGH [RED]
REASON : High coolant temp with elevated load suggests cooling system failure
ACTION : Stop vehicle immediately. Allow engine to cool. Check coolant level.

ANOMALOUS PARAMETERS:

- Engine Coolant Temperature: 135.64 C
- Engine Load: 108.32 %
- O2 Sensor Voltage: 0.03 V

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[RED] Mass Air Flow Rate
Normal value : 8.00 g/s
New value : 50.01 g/s (+525.1%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Short Term Fuel Trim: 0.00 %
OK Fuel Tank Pressure: 0.00 kPa
OK Throttle Position: 20.00 %
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Mass Air Flow Rate
Value: 50.01 g/s >> RED threshold (45.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------|-------|
| 1. Engine Overheating | 53.7% |
| 2. Air Intake Fault | 45.1% |
| 3. No Fault | 1.1% |

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FINAL DIAGNOSIS

=====

STATUS : ANOMALY DETECTED (fault type unclear)
ALERT : HIGH [RED]
CONFIDENCE : 53.7% (below threshold for fault ID)
ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Mass Air Flow Rate: 50.01 g/s

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 4 parameter(s)...

CHANGES MADE:

[ORANGE] Throttle Position
Normal value : 20.00 %
New value : 97.00 % (+385.0%)

[YELLOW] Battery Voltage
Normal value : 14.00 V
New value : 12.06 V (-13.9%)

[YELLOW] Mass Air Flow Rate
Normal value : 8.00 g/s
New value : 31.07 g/s (+288.4%)

[RED] Engine RPM
Normal value : 1500.00 rpm
New value : 5976.38 rpm (+298.4%)

Parameters NOT changed (remain normal):

- OK Engine Coolant Temperature: 92.00 C
- OK Engine Load: 35.00 %
- OK Intake Air Temperature: 25.00 C
- OK Short Term Fuel Trim: 0.00 %
- OK Fuel Tank Pressure: 0.00 kPa
- OK O2 Sensor Voltage: 0.45 V
- OK Vehicle Speed: 60.00 km/h
- OK Component Wear Index: 50.00 min

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DIAGNOSIS REPORT – RANDOM SIMULATION

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Engine RPM

Value: 5976.38 rpm >> RED threshold (5500.0)

[LOW [YELLOW]] Mass Air Flow Rate

Value: 31.07 g/s > YELLOW threshold (25.0)

[MEDIUM [ORANGE]] Throttle Position

Value: 97.00 % > ORANGE threshold (90.0)

[LOW [YELLOW]] Battery Voltage

Value: 12.06 V < YELLOW threshold (12.5)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------------|-------|
| 1. Air Intake Fault | 93.5% |
| 2. No Fault | 4.8% |
| 3. Engine Mechanical Stress | 1.4% |

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FINAL DIAGNOSIS

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STATUS : FAULT DETECTED

FAULT : Air Intake Fault

CONFIDENCE : 93.5%

ALERT : HIGH [RED]

REASON : Abnormal airflow and intake temperature suggest air system restriction

ACTION : Check air filter and throttle body. Inspect MAF sensor.

ANOMALOUS PARAMETERS:

- Engine RPM: 5976.38 rpm
 - Mass Air Flow Rate: 31.07 g/s
 - Throttle Position: 97.00 %
 - Battery Voltage: 12.06 V
- =====

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[ORANGE] Battery Voltage

Normal value : 14.00 V

New value : 11.62 V (-17.0%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C

OK Engine RPM: 1500.00 rpm

OK Engine Load: 35.00 %

OK Intake Air Temperature: 25.00 C

OK Short Term Fuel Trim: 0.00 %

OK Fuel Tank Pressure: 0.00 kPa

OK Mass Air Flow Rate: 8.00 g/s

OK Throttle Position: 20.00 %

OK O2 Sensor Voltage: 0.45 V

OK Vehicle Speed: 60.00 km/h

OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Battery Voltage

Value: 11.62 V < ORANGE threshold (12.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|---------------------------------|-------|
| 1. Electrical / Charging Fault | 98.8% |
| 2. Transmission / Speed Anomaly | 0.8% |
| 3. Engine Mechanical Stress | 0.3% |

FINAL DIAGNOSIS

STATUS : FAULT DETECTED

FAULT : Electrical / Charging Fault

CONFIDENCE : 98.8%

ALERT : MEDIUM [ORANGE]

REASON : Low battery voltage may indicate alternator or battery failure

ACTION : Test battery and alternator. Avoid turning off engine if possible.

ANOMALOUS PARAMETERS:

- Battery Voltage: 11.62 V

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[YELLOW] Short Term Fuel Trim
Normal value : 0.00 %
New value : -17.35 % (-Inf%)

Parameters NOT changed (remain normal):
OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Fuel Tank Pressure: 0.00 kPa
OK Mass Air Flow Rate: 8.00 g/s
OK Throttle Position: 20.00 %
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

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DIAGNOSIS REPORT – RANDOM SIMULATION

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[LOW [YELLOW]] Short Term Fuel Trim
Value: -17.35 % near boundary of normal

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:
1. No Fault 90.9%
2. Electrical / Charging Fault 3.8%
3. Engine Mechanical Stress 3.3%

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FINAL DIAGNOSIS

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STATUS : ANOMALY DETECTED
ALERT : LOW [YELLOW]
ACTION : Monitor affected parameters. Schedule inspection.

ANOMALOUS PARAMETERS:
- Short Term Fuel Trim: -17.35 %
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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)

2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 2 parameter(s)...

CHANGES MADE:

[RED] Engine Load

Normal value : 35.00 %

New value : 95.15 % (+171.9%)

[YELLOW] Vehicle Speed

Normal value : 60.00 km/h

New value : 158.55 km/h (+164.2%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C

OK Engine RPM: 1500.00 rpm

OK Intake Air Temperature: 25.00 C

OK Short Term Fuel Trim: 0.00 %

OK Fuel Tank Pressure: 0.00 kPa

OK Mass Air Flow Rate: 8.00 g/s

OK Throttle Position: 20.00 %

OK Battery Voltage: 14.00 V

OK O2 Sensor Voltage: 0.45 V

OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Engine Load

Value: 95.15 % >> RED threshold (95.0)

[LOW [YELLOW]] Vehicle Speed

Value: 158.55 km/h > YELLOW threshold (140.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------------|-------|
| 1. Engine Mechanical Stress | 56.0% |
| 2. Engine Overheating | 32.8% |
| 3. No Fault | 10.0% |

FINAL DIAGNOSIS

STATUS : ANOMALY DETECTED (fault type unclear)

ALERT : HIGH [RED]

CONFIDENCE : 56.0% (below threshold for fault ID)
ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Engine Load: 95.15 %
- Vehicle Speed: 158.55 km/h

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[ORANGE] Throttle Position
Normal value : 20.00 %
New value : 95.99 % (+380.0%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Short Term Fuel Trim: 0.00 %
OK Fuel Tank Pressure: 0.00 kPa
OK Mass Air Flow Rate: 8.00 g/s
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Throttle Position
Value: 95.99 % > ORANGE threshold (90.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------------|-------|
| 1. Air Intake Fault | 99.8% |
| 2. No Fault | 0.1% |
| 3. Engine Mechanical Stress | 0.1% |

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FINAL DIAGNOSIS

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STATUS : FAULT DETECTED
FAULT : Air Intake Fault
CONFIDENCE : 99.8%
ALERT : MEDIUM [ORANGE]
REASON : Abnormal airflow and intake temperature suggest air system restriction
ACTION : Check air filter and throttle body. Inspect MAF sensor.

ANOMALOUS PARAMETERS:

- Throttle Position: 95.99 %

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 3 parameter(s)...

CHANGES MADE:

[ORANGE] Short Term Fuel Trim
Normal value : 0.00 %
New value : -16.59 % (-Inf%)

[RED] Fuel Tank Pressure
Normal value : 0.00 kPa
New value : 1.71 kPa (+Inf%)

[ORANGE] O2 Sensor Voltage
Normal value : 0.45 V
New value : 0.03 V (-93.6%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Mass Air Flow Rate: 8.00 g/s
OK Throttle Position: 20.00 %
OK Battery Voltage: 14.00 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

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DIAGNOSIS REPORT – RANDOM SIMULATION

=====

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[LOW [YELLOW]] Short Term Fuel Trim
Value: -16.59 % near boundary of normal

[LOW [YELLOW]] Fuel Tank Pressure
Value: 1.71 kPa near boundary of normal

[MEDIUM [ORANGE]] O2 Sensor Voltage
Value: 0.03 V < ORANGE threshold (0.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|------------------------------|-------|
| 1. Fuel System Malfunction | 99.3% |
| 2. Exhaust / Emissions Fault | 0.3% |
| 3. No Fault | 0.2% |

FINAL DIAGNOSIS

STATUS : FAULT DETECTED
FAULT : Fuel System Malfunction
CONFIDENCE : 99.3%
ALERT : MEDIUM [ORANGE]
REASON : Abnormal fuel trim and O2 readings indicate fuel mixture problems
ACTION : Reduce speed. Schedule fuel injector or O2 sensor inspection.

ANOMALOUS PARAMETERS:

- Short Term Fuel Trim: -16.59 %
- Fuel Tank Pressure: 1.71 kPa
- O2 Sensor Voltage: 0.03 V

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 3 parameter(s)...

CHANGES MADE:

[RED] Mass Air Flow Rate
Normal value : 8.00 g/s
New value : 50.18 g/s (+527.2%)

[ORANGE] Component Wear Index
Normal value : 50.00 min

New value : 221.05 min (+342.1%)

[ORANGE] Intake Air Temperature

Normal value : 25.00 C

New value : 76.57 C (+206.3%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C

OK Engine RPM: 1500.00 rpm

OK Engine Load: 35.00 %

OK Short Term Fuel Trim: 0.00 %

OK Fuel Tank Pressure: 0.00 kPa

OK Throttle Position: 20.00 %

OK Battery Voltage: 14.00 V

OK O2 Sensor Voltage: 0.45 V

OK Vehicle Speed: 60.00 km/h

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DIAGNOSIS REPORT – RANDOM SIMULATION

=====

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Intake Air Temperature

Value: 76.57 C > ORANGE threshold (65.0)

[HIGH [RED]] Mass Air Flow Rate

Value: 50.18 g/s >> RED threshold (45.0)

[MEDIUM [ORANGE]] Component Wear Index

Value: 221.05 min > ORANGE threshold (200.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

1. Air Intake Fault	99.8%
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2. No Fault	0.1%
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3. Engine Overheating	0.1%
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FINAL DIAGNOSIS

=====

STATUS : FAULT DETECTED

FAULT : Air Intake Fault

CONFIDENCE : 99.8%

ALERT : HIGH [RED]

REASON : Abnormal airflow and intake temperature suggest air system restriction

ACTION : Check air filter and throttle body. Inspect MAF sensor.

ANOMALOUS PARAMETERS:

- Intake Air Temperature: 76.57 C

- Mass Air Flow Rate: 50.18 g/s

- Component Wear Index: 221.05 min

=====

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 2 parameter(s)...

CHANGES MADE:

[ORANGE] Battery Voltage

Normal value : 14.00 V

New value : 11.52 V (-17.7%)

[RED] Component Wear Index

Normal value : 50.00 min

New value : 251.50 min (+403.0%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C

OK Engine RPM: 1500.00 rpm

OK Engine Load: 35.00 %

OK Intake Air Temperature: 25.00 C

OK Short Term Fuel Trim: 0.00 %

OK Fuel Tank Pressure: 0.00 kPa

OK Mass Air Flow Rate: 8.00 g/s

OK Throttle Position: 20.00 %

OK O2 Sensor Voltage: 0.45 V

OK Vehicle Speed: 60.00 km/h

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DIAGNOSIS REPORT – RANDOM SIMULATION

=====

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Battery Voltage

Value: 11.52 V < ORANGE threshold (12.0)

[HIGH [RED]] Component Wear Index

Value: 251.50 min >> RED threshold (240.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

1. Engine Mechanical Stress	99.0%
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2. No Fault	0.6%
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3. Air Intake Fault	0.2%
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FINAL DIAGNOSIS

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STATUS : FAULT DETECTED

FAULT : Engine Mechanical Stress
CONFIDENCE : 99.0%
ALERT : HIGH [RED]
REASON : High RPM with excessive load indicates mechanical overstrain
ACTION : Reduce engine load. Avoid aggressive acceleration. Schedule inspection.

ANOMALOUS PARAMETERS:

- Battery Voltage: 11.52 V
- Component Wear Index: 251.50 min

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 1 parameter(s)...

CHANGES MADE:

[ORANGE] Throttle Position
Normal value : 20.00 %
New value : 93.05 % (+365.3%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Engine Load: 35.00 %
OK Intake Air Temperature: 25.00 C
OK Short Term Fuel Trim: 0.00 %
OK Fuel Tank Pressure: 0.00 kPa
OK Mass Air Flow Rate: 8.00 g/s
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

DIAGNOSIS REPORT – RANDOM SIMULATION

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Throttle Position
Value: 93.05 % > ORANGE threshold (90.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

1. Air Intake Fault	99.7%
2. No Fault	0.2%
3. Engine Mechanical Stress	0.1%

=====

FINAL DIAGNOSIS

=====

STATUS : FAULT DETECTED
FAULT : Air Intake Fault
CONFIDENCE : 99.7%
ALERT : MEDIUM [ORANGE]
REASON : Abnormal airflow and intake temperature suggest air system restriction
ACTION : Check air filter and throttle body. Inspect MAF sensor.

ANOMALOUS PARAMETERS:

- Throttle Position: 93.05 %

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 2

RANDOM SIMULATION

Starting state: ALL PARAMETERS NORMAL

Applying random disturbances to 3 parameter(s)...

CHANGES MADE:

[YELLOW] Fuel Tank Pressure
Normal value : 0.00 kPa
New value : 2.91 kPa (+Inf%)

[ORANGE] Mass Air Flow Rate
Normal value : 8.00 g/s
New value : 43.17 g/s (+439.6%)

[RED] Engine Load
Normal value : 35.00 %
New value : 102.34 % (+192.4%)

Parameters NOT changed (remain normal):

OK Engine Coolant Temperature: 92.00 C
OK Engine RPM: 1500.00 rpm
OK Intake Air Temperature: 25.00 C
OK Short Term Fuel Trim: 0.00 %
OK Throttle Position: 20.00 %
OK Battery Voltage: 14.00 V
OK O2 Sensor Voltage: 0.45 V
OK Vehicle Speed: 60.00 km/h
OK Component Wear Index: 50.00 min

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DIAGNOSIS REPORT – RANDOM SIMULATION
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LEVEL 1: Parameter Threshold Analysis (SAE J1979)
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```
[HIGH [RED]] Engine Load
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Value: 102.34 % >> RED threshold (95.0)
```

```
[HIGH [RED]] Fuel Tank Pressure
```

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Value: 2.91 kPa far outside normal [-1.0, 1.0]
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```
[MEDIUM [ORANGE]] Mass Air Flow Rate
```

```
Value: 43.17 g/s > ORANGE threshold (35.0)
```

```
LEVEL 2: Neural Network Fault Prediction (MLP)
-----
```

```
Top predicted faults:
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1. No Fault	46.5%
2. Engine Mechanical Stress	36.9%
3. Engine Overheating	14.2%

```
=====
FINAL DIAGNOSIS
=====
```

```
STATUS      : ANOMALY DETECTED (fault type unclear)
```

```
ALERT       : HIGH [RED]
```

```
CONFIDENCE  : 46.5% (below threshold for fault ID)
```

```
ACTION      : Inspect anomalous parameters. Seek professional diagnosis.
```

```
ANOMALOUS PARAMETERS:
```

- Engine Load: 102.34 %
- Fuel Tank Pressure: 2.91 kPa
- Mass Air Flow Rate: 43.17 g/s

```
-----
MAIN MENU
-----
```

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

```
-----
Select option (1/2/3): 3
```

```
System closed. Goodbye.
```

```
=====
>>
```